

AS Level Mathematics — Pure Mathematics 1 (P1)

Practice Paper 7

Time allowed: **1 hour 50 minutes**

Instructions to candidates

- Answer **all** questions. Write your answers on paper, showing all necessary working. Marks may be lost for insufficient working.
- If a numerical answer is not exact, give it correct to **3 significant figures**; give angles in degrees correct to **1 decimal place**, unless a different level of accuracy is specified in the question.
- A scientific calculator may be used. Do **not** use a graphical or symbolic calculator.
- The number of marks is given in brackets [] at the end of each question or part question. The total mark for this paper is **75**.
- This paper is written in the style of Cambridge International 9709 Paper 1 (Pure Mathematics 1). It is a practice paper, not an official examination paper.

List of formulae (as provided in the examination)

Quadratic equation: for $ax^2 + bx + c = 0$, $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Binomial expansion: $(a + b)^n = a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \dots + b^n$, where $\binom{n}{r} = \frac{n!}{r!(n-r)!}$

Arithmetic series: $u_n = a + (n-1)d$, $S_n = \frac{1}{2}n(a + l) = \frac{1}{2}n\{2a + (n-1)d\}$

Geometric series: $u_n = ar^{n-1}$, $S_n = \frac{a(1-r^n)}{1-r}$ ($r \neq 1$), $S_\infty = \frac{a}{1-r}$ ($|r| < 1$)

Circular measure: arc length $= r\theta$, sector area $= \frac{1}{2}r^2\theta$ (θ in radians)

Trigonometry: $\tan \theta \equiv \frac{\sin \theta}{\cos \theta}$, $\sin^2 \theta + \cos^2 \theta \equiv 1$

Differentiation and integration: $\frac{d}{dx}(x^n) = nx^{n-1}$, $\int x^n dx = \frac{x^{n+1}}{n+1} + c$ ($n \neq -1$)

1 Find the coefficient of x^4 in the expansion of $(3 - 2x)^7$. **[3]**

2 Solve the equation $4 \sin^2 x - 1 = 0$ for $0 \leq x \leq 2\pi$, giving your answers in terms of π . **[4]**

3 The second term of a geometric progression is 20 and the fifth term is 2.5. Find the first term, the common ratio and the sum to infinity of the progression. **[4]**

4

(a) Express $3x^2 + 12x + 5$ in the form $a(x + b)^2 + c$, where a , b and c are constants. **[3]**

(b) Hence solve the equation $3x^2 + 12x + 5 = 0$, giving your answers in exact form. **[2]**

5 A piece of wire of length 40 cm is bent to form the boundary of a sector of a circle with radius r cm and angle θ radians.

(a) Show that the area, A cm², of the sector is given by $A = 20r - r^2$. [2]

(b) Use differentiation to find the value of r for which A has a stationary value, and find the corresponding value of θ . [3]

(c) Determine whether this stationary value is a maximum or a minimum, and state the stationary value of A . [1]

6 The line l has equation $2y = x + 4$.

(a) Find the equation of the line through the point $P(3, -1)$ which is perpendicular to l , giving your answer in the form $y = mx + c$. [3]

(b) Hence find the coordinates of the foot of the perpendicular from P to l . [3]

7 The function f is defined by $f(x) = x^2 - 4x + 7$ for $x \leq 2$.

(a) Express $x^2 - 4x + 7$ in the form $(x - a)^2 + b$. [2]

(b) State the range of f . [1]

(c) Find an expression for $f^{-1}(x)$ and state the domain of f^{-1} . [3]

(d) Solve the equation $f(x) = 12$. [1]

8 A curve has equation $y = 2x^{\frac{3}{2}} - 9x + 1$ for $x > 0$.

(a) Find the coordinates of the stationary point of the curve and determine its nature. [4]

(b) Find the equation of the tangent to the curve at the point where $x = 4$. [3]

9

(a) Prove the identity $\frac{1 + \sin \theta}{\cos \theta} + \frac{\cos \theta}{1 + \sin \theta} \equiv \frac{2}{\cos \theta}$. [3]

(b) Hence solve the equation $\frac{1 + \sin \theta}{\cos \theta} + \frac{\cos \theta}{1 + \sin \theta} = 5$ for $0^\circ \leq \theta \leq 360^\circ$. [3]

(c) State the number of solutions of the equation $\sin 2x = 0.3$ in the interval $0^\circ \leq x \leq 360^\circ$. [2]

10

(a) Find the area of the region bounded by the curve $y = \frac{3}{\sqrt{2x+1}}$, the x -axis and the lines $x = 0$ and $x = 4$. [3]

(b) The region bounded by the curve $y = \sqrt{3x+1}$, the x -axis and the lines $x = 0$ and $x = 5$ is rotated through 360° about the x -axis. Find the volume of the solid formed, giving your answer in terms of π . [3]

(c) Evaluate $\int_1^4 \left(\frac{1}{\sqrt{x}} - 1 \right) dx$. [2]

11

(a) The sum of the first 10 terms of an arithmetic progression is 165, and the sum of the next 10 terms is 465.

(i) Find the first term and the common difference. [4]

(ii) Find the 25th term. [1]

(b) The sum of the first three terms of a geometric progression is 21 and the sum to infinity is 24. Find the common ratio and the first term. [4]

12 A circle has equation $x^2 + y^2 - 2x - 6y + 5 = 0$.

(a) Find the coordinates of the centre and the exact radius of the circle. [2]

(b) Show that the point $P(3, 4)$ lies on the circle, and find the equation of the tangent to the circle at P . [3]

(c) The tangent at P meets the x -axis at A and the y -axis at B . Find the area of triangle OAB , where O is the origin. [3]

END OF PAPER (Total: 75 marks)